Chapter 9, Problem 82P

Problem

A capacitance bridge balances when $R1$ = 100 Ω, $R2$ = 2 kΩ, and Cs = 40 μF. What is Cx, the capacitance of the capacitor under test?

Step-by-step solution

Step 1 of 1

$$C_X = \left(\frac{R_1}{R_2}\right)C_S$$

$$= \left(\frac{100}{2000}\right) \times (40\times10^{-6})$$

$$= 2\,\mu F \ .$$